OPENQUEST INNOVATION HACKATHON | DOMAIN 04

RAITU MITRA

రైతు మిత్ర • किसान मित्र

A farmer-first agricultural intelligence companion

Multilingual guidance • Crop health • Weather • Soil • Markets

MOBILE-FIRST | VOICE-FIRST | TRUST-FIRST

Agriculture & Food Systems | Telangana • Andhra Pradesh • India



The problem is not a lack of information

Farmers face fragmented, time-sensitive decisions across the crop cycle.

01 Late diagnosis

Disease and pest symptoms can look alike. Delayed or incorrect action can increase crop loss.

02 Scattered advice

Weather, soil, market and scheme information live across separate services.

03 Language barriers

Text-heavy, English-first tools are difficult to use in the field.

04 Unreliable connectivity

Farmers need useful guidance even when mobile data is weak or unavailable.



The opportunity: bring local, understandable and transparent decisions into one simple farmer journey.

One companion. Connected decisions.

Raitu Mitra brings essential farm signals into a single mobile experience.



See

Capture a crop image and receive an evidence-led health assessment.

Understand

Combine soil, weather and crop context into practical guidance.

Decide

Compare market information and plan the next farm action.

Access

Use Telugu, Hindi or English, with useful offline resources.

Designed around a real farmer workflow

A short, guided journey from question to next action.

01

Choose language

Telugu • Hindi • English

02

Capture / ask

Leaf photo, voice or typed question

03

Validate

Check image quality and plant relevance

04

Combine context

Weather + soil + crop + market

05

Act with confidence

Advice, source, timestamp or expert referral



Every recommendation should show what is known, what is estimated, and when expert help is needed.

A modular architecture built for field use

React PWA at the edge; FastAPI services and trusted data behind it.

FARMER EXPERIENCE

React + Vite PWA

- Mobile-first dashboard
- Telugu / Hindi / English
- Voice and camera
- Offline guides
- Farm map & records

INTELLIGENCE LAYER

FastAPI + Python

- Image validation
- Disease / pest services
- Weather & soil APIs
- Advisory engine
- Market proxy
- Chatbot & voice intent

DATA & MODELS

Evidence sources

- Plant disease datasets
- IMD / weather providers
- Soil maps & lab tests
- Agmarknet / eNAM
- Satellite products
- Farmer-entered context

Security boundary: secrets stay on the server • data sources carry dates • uncertain results are not presented as facts

Responsible crop diagnosis

A validation-first pipeline prevents unrelated photos from becoming disease claims.

1 Image checks

File type • resolution • blur

2 Plant validation

Plant / non-plant / uncertain

3 Disease model

38 PlantVillage classes (reported)

4 Explanation

Prediction + confidence + Grad-CAM

5 Safe next step

Guidance or expert escalation



Reported internal held-out PlantVillage accuracy: 87.63% — not yet evidence of real-farm accuracy.

From isolated signals to useful advice

The advisory engine should combine context—not just display separate dashboards.

WEATHER

Current conditions
7-day forecast
Rain / wind alerts

SOIL

Lab-test values
Regional soil context
N • P • K • pH • EC

SATELLITE

Field boundary
Vegetation indices
Acquisition date & resolution

MARKET

Min / max / modal
Market & commodity
Date + source

RAITU MITRA ADVISORY ENGINE

Crop stage + location + farmer inputs + verified data → explainable next action

A regional estimate must never be shown as a laboratory measurement; MSP must never be shown as a live mandi quote.

Voice-first, multilingual and accessible

Designed for farmers who prefer speaking over typing.

VOICE NAVIGATOR

Speech-to-text, intent detection and guided navigation.

MITRA CHAT

Agriculture-focused questions with safe, grounded responses.

3 LANGUAGES

Telugu • Hindi • English, with text alternatives.

EXPERT ESCALATION

Uncertain or high-risk cases should route to an agricultural officer / KVK.



Voice is an access layer—not a replacement for evidence, consent or farmer confirmation.

Trust is a product feature

Agricultural advice can affect a farmer’s income. Uncertainty must be visible.

No fake data

Never invent prices, weather, soil values or model outputs.

Know the source

Show provider, observation date, location and spatial scale.

Show uncertainty

Low confidence → explain limits and suggest expert review.

Protect the farmer

Secure API keys, uploads, location and personal records.

Safe recommendations

Prefer IPM and verified labels; avoid unsupported chemical dosage.

Build snapshot: honest status, clear next steps

Based on the supplied codebase audit; each item still requires repeatable verification.

MODULE	AUDIT SNAPSHOT	COMPLETION FOCUS
Disease classifier	Model artifact reported present	Validate on real-field images
Plant-image gate	Partially implemented	Harden non-plant rejection
Pest detection	Missing dedicated model	Dataset + model + evaluation
Weather	Partially implemented	Secure backend + GPS + forecast
Soil	Partially implemented	Regional provenance + lab-test workflow
Satellite	Module reported present	Verify live source, dates & resolution
Mandi	Partially implemented	Backend proxy + dated official data
Voice / chatbot	Reported implemented	Multilingual end-to-end tests
PWA / offline	Partially implemented	Install + cache + offline tests
Deployment	In progress	Deploy and test public URLs

Measure impact, not just features

A credible pilot should track outcomes that matter to farmers.

Faster triage

Time from photo to useful next step

Better decisions

Share of advice with source + timestamp

Accessible use

Task completion in Telugu / Hindi / English

Field reliability

Success rate on low-end Android + weak networks

Responsible AI

False-accept rate on non-plant images; uncertain-case handling



Pilot with farmers, extension workers and KVK experts before scaling.

LIVE DEMO • 5–7 MINUTES

From a farmer’s question to a safer decision

1. Switch to Telugu and ask by voice.
2. Upload a genuine leaf image; show validation and diagnosis.
3. Open weather and explain source + forecast timestamp.
4. Enter soil-test values and review interpretation.
5. Compare a dated mandi quote with MSP separately.
6. Show field/satellite metadata and offline guides.

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Local language. Connected intelligence. Honest advice.

Thank you

